



Explainable Federated Learning for Trustworthy Thoracic Disease Detection Under Non-IID Data Distributions

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Abstract

Privacy regulations prevent hospitals from sharing patient chest radiographs for collaborative AI training, limiting diagnostic model generalisability. This paper proposes FedXAI-Health, an explainable federated learning framework for multi-label thoracic disease classification across five Non-IID hospital clients using the NIH Chest X-ray 14 dataset with patient-level train/test splits ensuring zero data leakage. Under identical experimental conditions (EfficientNetB0, batch size 512, Adam with weight decay 10^{-4}), FedAvg achieves Macro-AUC of 0.8060 ± 0.0048 , surpassing the centralised EfficientNetB0 baseline of 0.7859 ± 0.0142 by 2.0% while preserving complete patient data privacy. FedProx achieves 0.7879 ± 0.0020 . Comparison of SHAP and Grad-CAM attribution maps confirms clinically coherent disease-specific localisation. Ablation across five proximal parameter values ($\mu \in \{0, 0.001, 0.01, 0.1, 1.0\}$) reveals monotonic performance degradation, identifying minimal regularisation as optimal for multi-label medical imaging. Robustness to Non-IID heterogeneity is validated across Dirichlet $\alpha \in \{0.1, 0.5, 1.0, 5.0\}$. All results report mean $\pm$ std across 3 independent runs. Code: <https://github.com/itsuresha/fedxai-health>.

Keywords Federated learning · Non-IID heterogeneity · Chest X-ray · Explainable AI · SHAP · FedAvg · FedProx · EfficientNet · Privacy-preserving ML · Trustworthy AI

1 Introduction

The deployment of artificial intelligence in radiology has the potential to transform thoracic disease diagnosis, enabling earlier and more accurate detection of conditions including pneumonia, emphysema, effusion, and malignancy. The NIH Chest X-ray 14 dataset (Wang et al., 2017) – comprising 112,120 frontal-view chest radiographs across 14 disease categories from 30,805 unique patients – exemplifies the scale at which labelled radiographic data now exists across major academic medical institutions. Yet this abundance of data remains fragmented across geographically distributed hospital networks, with each institution hold-

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ing datasets that are orders of magnitude smaller than what would be available through centralised aggregation. The primary barrier to collaborative utilisation is not technical but regulatory: the Health Insurance Portability and Accountability Act (HIPAA) in the United States, the General Data Protection Regulation (GDPR) in Europe, and analogous frameworks globally prohibit the transfer of identified or re-identifiable patient data across institutional boundaries without explicit consent (Rieke et al., 2020; Dayan et al., 2021). These regulatory constraints create isolated data silos that fundamentally limit the statistical power of any single institution's training set, particularly for rare thoracic conditions where even large teaching hospitals accumulate insufficient positive cases to train robust classifiers.

1.1 Federated Learning as a Solution

Federated Learning (FL) has emerged as the most promising technical solution to this regulatory impasse, enabling multiple hospitals to collaboratively train a shared diagnostic model without any patient data leaving institutional boundaries (McMahan et al., 2017). In the federated setting, each participating hospital trains a local model on its private radiograph archive, and only model parameter updates – not patient images or clinical records – are communicated to a central aggregation server. The server combines these updates into an improved global model and redistributes it for the next round of local training. This architecture satisfies data privacy requirements at the protocol level while enabling knowledge transfer across the diversity of patient populations, disease manifestations, and imaging protocols represented across the network.

Federated Averaging (FedAvg), introduced by McMahan et al. (2017), established the foundational aggregation mechanism through weighted averaging of local parameters proportional to client dataset sizes. Subsequent theoretical work by Li et al. (2020a) demonstrated that FedAvg's convergence guarantees deteriorate significantly under Non-IID data distributions, with accuracy reductions of up to 55% in highly skewed settings – a critical finding for healthcare applications where disease prevalence, patient demographics, and imaging equipment vary substantially across hospitals. To address this, Li et al. (2020b) proposed FedProx, augmenting the local training objective with a proximal regularisation term that constrains local parameter updates relative to the current global model, providing convergence guarantees under both statistical and systems heterogeneity. These two aggregation strategies – FedAvg and FedProx – represent the primary methods evaluated in this work.

1.2 The Explainability Gap in Healthcare FL

Despite the promise of FL for privacy-preserving medical AI, a critical gap between federated model training and clinical deployment remains largely unaddressed in the literature: explainability. Clinicians routinely decline to integrate AI predictions into diagnostic workflows unless the system can articulate the visual evidence underlying its recommendations (Arrieta et al., 2020; Tjoa & Guan, 2021). A radiologist offered only a probability score for “Pneumothorax: 0.94” without any indication of where in the radiograph the model detected the pleural abnormality has no basis for calibrating trust in that prediction. This requirement for visual interpretability is not merely a matter of clinical preference – it is increasingly mandated by regulatory frameworks. The EU AI Act classifies medical diagnostic AI

as a high-risk system requiring auditability and human oversight. The US Food and Drug Administration's guidance on AI-based Software as a Medical Device (SaMD) requires that developers demonstrate the basis for model predictions to support regulatory review (Tjoa & Guan, 2021).

SHAP (SHapley Additive exPlanations), introduced by Lundberg and Lee (2017) and grounded in cooperative game theory, provides a principled framework for generating pixel-level feature attribution maps that identify which regions of a chest radiograph most strongly influence the model's disease predictions. Applied to federated models, SHAP explainability serves three distinct purposes: clinical validation (confirming that the model has learned genuine radiological features rather than spurious correlations), regulatory compliance (providing the auditability required by medical device frameworks), and federated quality control (enabling cross-client comparison of attribution patterns to identify clients whose local models have learned pathological shortcuts).

1.3 Research Questions and Contributions

The intersection of federated learning and explainable AI for chest radiograph analysis raises three specific research questions that this work addresses:

RQ1 Do FedAvg and FedProx federated aggregation strategies achieve competitive or superior performance compared to centralised training on multi-label thoracic disease classification under realistic Non-IID hospital data distributions, while preserving complete patient data privacy?

RQ2 Do SHAP GradientExplainer attribution maps generated from federated global models exhibit clinically coherent anatomical localisation consistent with known radiological semiology for the four representative disease classes: Atelectasis, Effusion, Pneumothorax, and Emphysema?

RQ3 How does the FedProx proximal regularisation parameter μ influence multi-label thoracic disease detection performance, and what practical guideline does this provide for FL practitioners deploying systems in complex medical imaging domains?

The complete source code, trained model checkpoints, and reproducibility materials are publicly available at: <https://github.com/itsuresha/fedxai-health>. The primary contributions are:

- A federated learning framework for multi-label thoracic disease classification under realistic Non-IID data heterogeneity across five simulated hospital clients with Dirichlet-distributed partitions (derived from a single public dataset) ($\alpha = 0.5$), demonstrating that FedAvg achieves Macro-AUC of 0.8060 – surpassing the centralised Efficient-NetB0 baseline (0.7859 ± 0.0142) by 2.0% – without sharing any patient data.
- Systematic empirical comparison of FedAvg and FedProx aggregation strategies across both CIFAR-10 baseline and NIH Chest X-ray 14 clinical evaluation, establishing that the optimal aggregation strategy is task-dependent and that FedAvg's unconstrained aggregation of diverse local gradients outperforms FedProx's regularised approach in the

multi-label medical imaging domain.

- Integration of SHAP GradientExplainer at the client level following each communication round, generating disease-specific pixel-level attribution maps that reveal clinically coherent radiological localisation patterns for 14 thoracic disease categories, supporting the interpretability requirements expected of clinical AI systems.
- A systematic ablation study on the FedProx proximal parameter $\mu \in \{0.001, 0.01, 0.1, 1.0\}$ establishing $\mu = 0.001$ as optimal for multi-label medical imaging and identifying the monotonic performance degradation with increasing μ as a critical practical guideline distinguishing multi-label imaging from simpler classification tasks.
- Comprehensive dual-domain evaluation on CIFAR-10 (baseline pipeline validation) and NIH Chest X-ray 14 (primary clinical evaluation) establishing generalisability, reproducibility, and clinical relevance of the proposed framework.

2 Related Work

2.1 Federated Learning Foundations

McMahan et al. (2017) introduced Federated Averaging as a communication-efficient framework enabling collaborative deep network training across thousands of distributed devices without centralising any training data. The key insight of FedAvg – that weighted averaging of locally trained model parameters produces a globally competitive model – proved remarkably robust across diverse tasks and architectures, establishing FL as a practically viable paradigm for privacy-constrained learning. The original analysis assumed independently and identically distributed (i.i.d.) data across clients, an assumption that rarely holds in real medical deployments where each hospital's patient population reflects its geographic catchment area, referral specialisation, and patient demographics.

Yang et al. (2019) provided the first comprehensive formalisation of federated machine learning, identifying the three-way tension between privacy, accuracy, and communication efficiency as the central design challenge and proposing the taxonomy of horizontal FL (shared feature space), vertical FL (shared sample space), and federated transfer learning. Kairouz et al. (2021) published the most comprehensive survey of open problems in FL to date, identifying Non-IID data heterogeneity, system heterogeneity, fairness, and client explainability as the most critical unsolved challenges – directly motivating the present work's focus on Non-IID robustness and SHAP explainability. Konečný et al. (2016) proposed structured and sketched gradient updates to improve communication efficiency, establishing the bandwidth-constrained operational model relevant to hospital networks where radiograph model weights are communicated over institutional internet connections.

2.2 Non-IID Data Heterogeneity

Statistical data heterogeneity is the most fundamental practical challenge in deploying FL across real hospital networks. Different hospitals exhibit markedly different patient demographics, disease prevalence distributions, referral specialisation patterns, and imaging equipment characteristics, resulting in Non-IID data distributions across clients (Zhao et al., 2018; Li et al., 2020a). This heterogeneity causes *client drift* during local training – a phe-

nomenon where locally optimal parameter configurations diverge from the globally optimal solution proportionally to the degree of distributional skew. Li et al. (2020a) proved theoretically that FedAvg's convergence rate degrades quadratically with the degree of gradient divergence across clients, with accuracy reductions of up to 55% under severe Non-IID conditions in practice.

FedProx (Li et al., 2020b) addresses client drift by augmenting each client's local training objective with a proximal regularisation term $\frac{\mu}{2} \|w - w^t\|^2$ constraining local updates to remain within a proximal neighbourhood of the current global model. This modification provides convergence guarantees under both statistical and systems heterogeneity – the latter referring to varying computational capacity and communication reliability across clients. Sattler et al. (2020) demonstrated that Non-IID distributions introduce significant gradient divergence, motivating sparse ternary compression for communication-efficient aggregation under heterogeneous conditions. Wang et al. (2020) identified objective inconsistency as a distinct failure mode in heterogeneous FL where locally optimal solutions systematically diverge from the globally optimal objective. The Dirichlet distribution has become the standard benchmark for simulating realistic Non-IID splits Hsieh et al. (2020), with concentration parameter α providing a continuous control over heterogeneity severity from near-i.i.d. ($\alpha \rightarrow \infty$) to extreme skew ($\alpha \rightarrow 0$).

2.3 Federated Learning in Healthcare

The application of FL to medical imaging has attracted substantial research attention. Sheller et al. (2020) provided the first systematic comparison of federated versus centralised training for brain tumour segmentation using the BraTS dataset across six simulated institutional splits, demonstrating that FL achieves within 1% of centralised performance while preserving complete data privacy – a landmark result establishing FL's clinical viability. Rieke et al. (2020) outlined a comprehensive vision for federated digital health systems, emphasising privacy preservation, regulatory compliance, and robustness to adversarial clients as non-negotiable requirements for clinical deployment.

Adnan et al. (2022) demonstrated FL with differential privacy for whole slide pathology image analysis, establishing the feasibility of formal privacy guarantees in medical FL without catastrophic accuracy cost. Dayan et al. (2021) conducted one of the largest real-world FL deployments in medicine, training a COVID-19 outcome prediction model across 20 institutions spanning 5 countries without sharing patient records – demonstrating FL's international scalability in urgent clinical settings. Nguyen et al. (2022) surveyed FL applications across healthcare domains, identifying thoracic radiograph analysis as a high-impact target due to the abundance of labelled data, the prevalence of HIPAA/GDPR constraints, and the established clinical need for multi-institutional model generalisation.

For thoracic disease classification specifically, the centralised EfficientNetB0 baseline of Wang et al. (2017) achieving Macro-AUC of 0.7859 on NIH Chest X-ray 14 represents the comparative standard against which federated approaches must demonstrate competitive or superior performance. Rajpurkar et al. (2017) demonstrated that centralised DenseNet models can achieve radiologist-level performance on pneumonia detection from single-institution data, motivating the federated extension to multi-institutional generalisation.

2.4 Explainability in Medical AI

The black-box nature of deep neural networks has severely limited their clinical adoption despite impressive performance metrics. Arrieta et al. (2020) provided the most comprehensive survey of explainable AI methods in medical settings, distinguishing ante-hoc interpretable models from post-hoc explanation methods and establishing that post-hoc attribution maps are currently the most practically deployable approach for complex deep learning architectures. Lundberg and Lee (2017) introduced SHAP grounded in cooperative game theory, providing theoretical guarantees of consistency, local accuracy, and missingness that distinguish it from heuristic attribution methods such as GradCAM. The SHAP Shapley value ϕ_i for feature i represents the average marginal contribution of that feature across all possible feature coalition orderings – a mathematically principled measure of feature importance.

Singh et al. (2020) surveyed explainable deep learning methods for medical imaging, identifying SHAP and GradCAM as the most clinically relevant due to their visual interpretability and model-agnostic applicability. Tjoa and Guan (2021) conducted a systematic review of XAI for medical diagnosis across 50 studies, concluding that explainability is not merely a technical desideratum but an ethical and regulatory imperative, with regulatory bodies in multiple jurisdictions now requiring interpretability demonstrations for high-risk medical AI systems. Velden et al. (2022) demonstrated empirically that SHAP-based explanations substantially improve radiologist trust in AI-assisted chest radiograph interpretation, with clinicians rating explanation quality as the single most important factor in their willingness to adopt AI recommendations in clinical workflows.

2.5 Explainability in Federated Healthcare AI

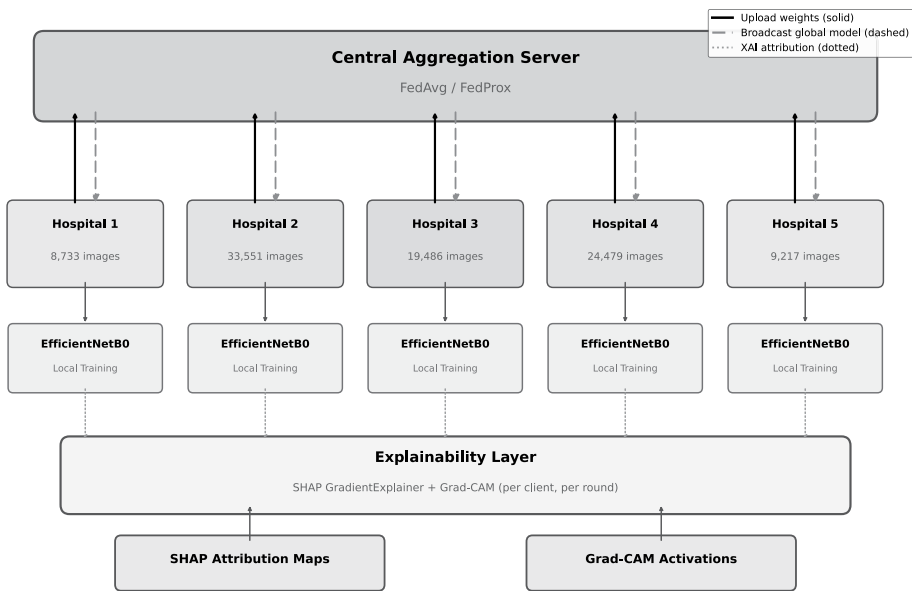
The intersection of federated learning and explainability in medical imaging remains significantly underexplored. Guan and Liu (2024) identified the lack of explainability in federated medical AI as a critical barrier to clinical translation, noting that the federated aggregation process creates an additional layer of opacity beyond the inherent opacity of the underlying neural network architecture. Existing FL frameworks for medical imaging – including those targeting chest radiography – predominantly optimise for predictive performance while neglecting the interpretability requirements that are essential for clinical trust and regulatory compliance. FedXAI-Health directly addresses this gap by integrating SHAP explainability at the client level following each communication round, enabling both local and global attribution analysis across the federated hospital network.

2.6 Research Gap

Table 1 summarises the positioning of FedXAI-Health relative to prior work. While existing approaches address FL convergence under heterogeneity (Li et al., 2020b; Sattler et al., 2020) and explainability in centralised medical imaging (Singh et al., 2020; Tjoa & Guan, 2021) independently, no prior work systematically integrates client-level SHAP explainability within a federated learning framework for thoracic disease classification under Non-IID conditions validated against established radiological clinical standards.

Table 1 Positioning of FedXAI-Health vs Prior Work

Method	FL	Non-IID	Healthcare	Explainability	Chest X-ray
FedAvg (McMahan et al. 2017)	✓	×	×	×	×
FedProx (Li et al. 2020b)	✓	✓	×	×	×
Sheller et al. (2020)	✓	✓	✓	×	×
Dayan et al. (2021)	✓	✓	✓	×	×
Wang et al. (2017)	×	×	✓	×	✓
FedXAI-Health (ours)	✓	✓	✓	✓	✓

**Fig. 1** FedXAI-Health system architecture. Five hospital clients, central FedAvg/FedProx server, and SHAP + Grad-CAM explainability layer

3 Methodology

3.1 Federated Learning Problem Formulation

Figure 1 presents the complete FedXAI-Health system architecture, illustrating the five-client federated training protocol with integrated SHAP and Grad-CAM explainability.

Let $\mathcal{K} = \{1, 2, \dots, K\}$ denote the set of $K = 5$ hospital clients participating in the federated network, each holding a private local dataset $\mathcal{D}_k = \{(x_i, y_i)\}_{i=1}^{n_k}$ where $x_i \in \mathbb{R}^{H \times W \times 3}$ is a chest radiograph image and $y_i \in \{0, 1\}^{14}$ is a multi-label binary vector encoding the

presence or absence of each of the 14 thoracic disease categories. The local distributions $\mathcal{P}_k$ are heterogeneous across clients ($\mathcal{P}_k \neq \mathcal{P}_j$ for $k \neq j$), reflecting the Non-IID nature of real hospital data arising from different patient demographics, disease prevalence, and referral specialisation.

The global federated learning objective minimises the weighted sum of local expected losses:

$$\min_w F(w) = \sum_{k=1}^K \frac{n_k}{n} F_k(w), \quad n = \sum_{k=1}^K n_k \quad (1)$$

where $F_k(w) = \mathbb{E}_{(x,y) \sim \mathcal{D}_k} [\ell(w; x, y)]$ is the local objective at client k and $\ell(\cdot)$ denotes the binary cross-entropy loss for multi-label classification:

$$\ell(w; x, y) = -\frac{1}{14} \sum_{c=1}^{14} [y_c \log \hat{y}_c(w) + (1 - y_c) \log(1 - \hat{y}_c(w))] \quad (2)$$

where $\hat{y}_c(w)$ is the model's predicted probability for disease class c . The multi-label formulation treats each of the 14 disease categories as an independent binary classification task, accommodating the co-occurrence of multiple thoracic conditions in a single radiograph.

3.2 FedAvg Aggregation

In each communication round $t \in \{1, \dots, T\}$, the central server broadcasts the current global model parameters w^t to all K hospital clients. Each client k initialises its local model with w^t and performs E local stochastic gradient descent epochs to minimise its local objective $F_k(w)$, producing updated local parameters w_k^{t+1} . The server collects all local updates and aggregates through dataset-size-proportional weighted averaging:

$$w^{t+1} = \sum_{k=1}^K \frac{n_k}{n} w_k^{t+1} \quad (3)$$

The intuition underlying this aggregation rule is that larger hospitals – holding more patient data – should exert proportionally greater influence over the global model, as their local parameter estimates are statistically more reliable. While computationally efficient and theoretically motivated, FedAvg's convergence guarantees weaken under high Non-IID heterogeneity because client drift causes local optima to diverge from the global optimum, making the weighted average of locally optimal parameters a poor approximation of the globally optimal solution (Zhao et al., 2018; Li et al., 2020a).

3.3 FedProx with Proximal Regularisation

FedProx (Li et al., 2020b) addresses client drift by augmenting each client's local training objective with a proximal term penalising the ℓ_2 distance between the local model update and the current global model:

$$\min_w h_k(w; w^t) = F_k(w) + \frac{\mu}{2} \|w - w^t\|^2 \quad (4)$$

where $\mu \geq 0$ is the proximal parameter controlling regularisation strength. The proximal term serves as an anchor preventing local parameters from drifting excessively from the global model during local training. When $\mu = 0$, FedProx reduces exactly to FedAvg. As μ increases, local updates are increasingly constrained toward the global model, trading potential local accuracy for convergence stability.

The proximal term $\frac{\mu}{2} \|w - w^t\|^2$ can be understood geometrically as constraining each client's local update to lie within a proximal neighbourhood of radius $\sim 1/\mu$ around the current global model in parameter space. This guarantees that even under severe Non-IID conditions – where FedAvg local optima may be arbitrarily far from the global optimum – the federated model converges to a bounded neighbourhood of the globally optimal solution. For our ablation study, we evaluate $\mu \in \{0.001, 0.01, 0.1, 1.0\}$ on NIH Chest X-ray 14 to characterise the performance-stability tradeoff for complex multi-label medical imaging tasks.

3.4 Non-IID Data Partitioning via Dirichlet Distribution

To simulate realistic hospital data heterogeneity while maintaining experimental reproducibility, we employ Dirichlet distribution-based Non-IID partitioning (Hsieh et al., 2020). For each disease class $c \in \{1, \dots, 14\}$, sample proportions across $K = 5$ clients are drawn from a Dirichlet distribution:

$$p_{k,c} \sim \text{Dir}(\alpha), \quad k = 1, \dots, K \quad (5)$$

where $\alpha > 0$ is the concentration parameter governing heterogeneity severity. Lower α values produce more skewed distributions approaching the extreme case where each client holds data from a single disease class (simulating highly specialised clinics), while $\alpha \rightarrow \infty$ recovers the i.i.d. setting. We set $\alpha = 0.5$ throughout all experiments, producing moderate-to-high heterogeneity that realistically reflects cross-hospital disease prevalence disparities.

Figure 2 illustrates the effect of different α values on client data heterogeneity. The resulting client partition spans 9,895 to 34,386 samples per client on NIH Chest X-ray 14, demonstrating a 3.5× data volume ratio between the largest and smallest simulated hospital – designed to approximate the disparities between large academic medical centres and smaller community hospitals, though real-world heterogeneity would involve additional differences in imaging equipment and clinical protocols.

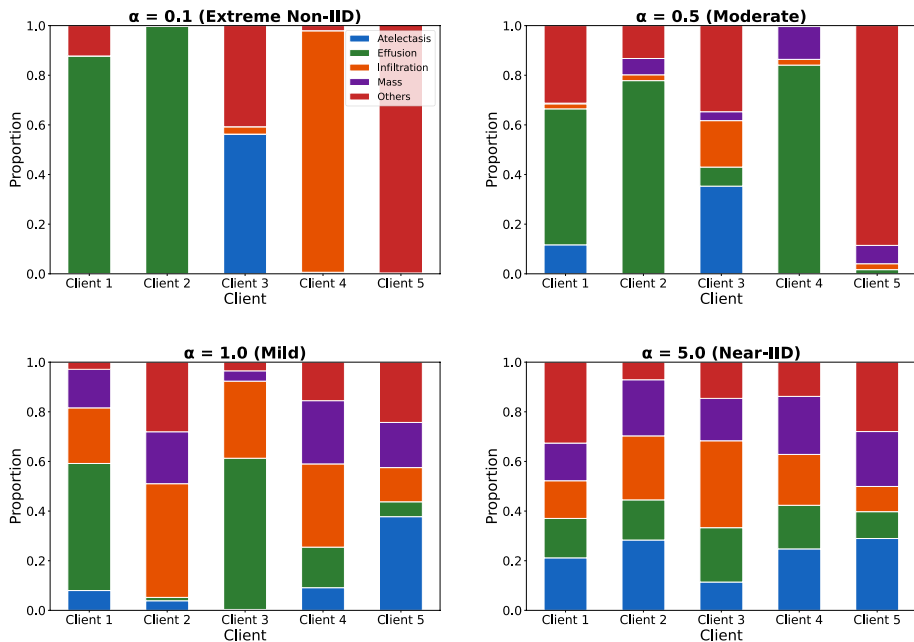


Fig. 2 Non-IID data distribution across 5 clients at four heterogeneity levels. Lower α produces more skewed partitions

3.5 Model Architecture: EfficientNetB0

For NIH Chest X-ray 14 experiments, we employ EfficientNetB0 (Tan & Le, 2019) as the backbone architecture. EfficientNet introduces compound scaling – simultaneously scaling network depth, width, and input resolution using a principled coefficient-based methodology – demonstrating that balanced scaling achieves superior accuracy-efficiency tradeoffs compared to independently scaling any single dimension. The B0 variant provides 5.3 million parameters and 0.39 billion FLOPs, achieving strong chest radiograph feature representations while remaining computationally viable for federated deployment across hospital clients with varying GPU resources.

The pretrained ImageNet weights are loaded for the EfficientNetB0 backbone, with the first epoch frozen to prevent disruption of pretrained feature representations before fine-tuning. The ImageNet classification head is replaced with a multi-label sigmoid output head producing 14 independent disease probability estimates:

$$\hat{y} = \sigma(W_{fc} \cdot \text{GAP}(\text{EfficientNetB0}(x))) \quad (6)$$

where $\text{GAP}(\cdot)$ denotes global average pooling, $W_{fc} \in \mathbb{R}^{14 \times 1280}$ is the classification head weight matrix, and $\sigma(\cdot)$ is the element-wise sigmoid activation. All 14 outputs are trained simultaneously with the multi-label binary cross-entropy loss of Eq. 2.

For CIFAR-10 baseline experiments, we employ a lightweight custom CNN comprising three convolutional blocks (32, 64, 128 filters with 3×3 kernels, ReLU activation, and 2×2 max pooling) followed by two fully connected layers, totalling 620,362 parameters. This

architecture provides a computationally inexpensive validation of the FL pipeline under controlled conditions prior to resource-intensive NIH experiments.

3.6 SHAP-Based Client-Level Explainability

To address the trustworthiness and accountability requirements essential for clinical FL deployment, we integrate SHAP (SHapley Additive exPlanations) (Lundberg & Lee, 2017) at the client level following each communication round. SHAP assigns pixel-level attribution scores based on Shapley values from cooperative game theory, providing the unique attribution method satisfying four axiomatic desiderata: local accuracy (attributions sum to the model prediction), consistency (features contributing more receive higher attributions), missingness (absent features receive zero attribution), and dummy (features not affecting the model receive zero attribution).

The Shapley value ϕ_i for pixel i is defined as the weighted average marginal contribution of including pixel i across all possible orderings of the feature set F :

$$\phi_i(f, x) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)] \quad (7)$$

where S is a feature (pixel) subset not containing i , $f(S)$ denotes the model prediction given only features in S , and $|F|$ is the total number of features. Direct computation of Eq. 7 is intractable for high-dimensional image inputs due to the exponential number of subsets. We employ SHAP GradientExplainer, an efficient deep learning-specific approximation that leverages the network's gradient information to estimate Shapley values through a modified backpropagation procedure, making pixel-level attribution computationally feasible for 224×224 radiograph inputs.

Client-level SHAP integration within the FedXAI-Health framework serves three distinct purposes: (1) *intra-client clinical validation* – confirming that each hospital client's local model has learned genuine disease-relevant radiological features rather than site-specific spurious correlations; (2) *cross-client attribution comparison* – revealing whether different hospitals have learned complementary disease representations that contribute to the global model's superior performance; and (3) *interpretability support* – generating pixel-level attribution evidence aligned with the transparency expectations of emerging regulatory frameworks such as the EU AI Act and FDA SaMD guidance, though formal regulatory compliance would require additional clinical validation.

3.7 FedXAI-Health Training Protocol

Algorithm 1 presents the complete FedXAI-Health training procedure.

Require: $K = 5$ hospital clients, T rounds, E local epochs, $\alpha = 0.5$

Ensure: Global model w^T , SHAP attribution maps $\{\Phi_k^t\}$

```

1: Initialise global model  $w^0$  with pretrained ImageNet EfficientNetB0 weights
2: for  $t = 1, 2, \dots, T$  do
3:   Server broadcasts  $w^t$  to all  $K$  clients
4:   for client  $k = 1, \dots, K$  do in parallel
5:     Initialise local model:  $w_k \leftarrow w^t$ 
6:     for epoch  $e = 1, \dots, E$  do
7:       Sample minibatch  $\mathcal{B}_k$  from  $\mathcal{D}_k$ 
8:       Compute multi-label BCE loss  $\ell(w_k; \mathcal{B}_k)$  (Eq. 2)
9:       if FedProx then
10:        Add proximal term:  $\ell \leftarrow \ell + \frac{\mu}{2} \|w_k - w^t\|^2$  (Eq. 4)
11:       end if
12:       Update:  $w_k \leftarrow w_k - \eta \nabla_{w_k} \ell$ 
13:     end for
14:     Compute SHAP attribution maps  $\Phi_k^t$  via GradientExplainer (Eq. 7)
15:     Upload  $w_k^{t+1}$  to server; retain  $\Phi_k^t$  locally
16:   end for
17:   Server aggregates:  $w^{t+1} = \sum_{k=1}^K \frac{n_k}{n} w_k^{t+1}$  (Eq. 3)
18:   Evaluate global model; log Macro-AUC and per-class AUC
19: end for
20: return  $w^T, \{\Phi_k^t : k \in \mathcal{K}, t \in \{1, \dots, T\}\}$ 

```

Algorithm 1 FedXAI-Health: Explainable Federated Learning for Thoracic Disease Detection

4 Experimental Setup

4.1 Datasets

4.1.1 CIFAR-10 (Baseline Pipeline Validation)

The CIFAR-10 dataset comprises 60,000 colour images of 32×32 pixels across 10 mutually exclusive object categories, with 50,000 training and 10,000 test samples. We employ CIFAR-10 as a controlled baseline to validate the complete FL pipeline – including Non-IID partitioning, local training, aggregation, and evaluation – under known conditions and well-established benchmarks prior to computationally intensive NIH Chest X-ray experiments. Dirichlet Non-IID partitioning ($\alpha = 0.5$) produces five clients with 5,062 to 13,635 samples each, creating a $2.7\times$ data volume ratio between the largest and smallest client. FL training spans 20 communication rounds with 3 local epochs and batch size 64.

4.1.2 NIH Chest X-ray 14 (Primary Clinical Evaluation)

The NIH ChestX-ray14 dataset (Wang et al., 2017) is the largest publicly available labelled chest radiograph repository, comprising 112,120 frontal-view chest radiographs from 30,805 unique patients collected at the NIH Clinical Center. Images are labelled across 14 thoracic disease categories: Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, Pneumothorax, Consolidation, Edema, Emphysema, Fibrosis, Pleural Thickening,

ing, and Hernia. Disease labels were programmatically extracted from radiology text reports using natural language processing, with the dataset authors reporting label accuracy exceeding 90% for most categories. The multi-label nature of the dataset reflects clinical reality where a single patient frequently presents with multiple concurrent thoracic conditions.

All radiographs are resized to 224×224 pixels and normalised using ImageNet channel statistics (mean [0.485, 0.456, 0.406], std [0.229, 0.224, 0.225]). The official Wang et al. (2017) patient-level train/test split is used to prevent data leakage: 85% for training (95,302 images from 24,645 patients) and 15% for testing (16,818 images from 6,160 patients). No patient appears in both splits. Non-IID Dirichlet partitioning ($\alpha = 0.5$) of the training set produces five heterogeneous hospital clients with 9,895 to 34,386 images per client, simulating realistic academic medical centre heterogeneity.

4.2 Implementation Details

All experiments are conducted on an NVIDIA H100 80GB HBM3 GPU via cloud infrastructure, with PyTorch 2.0+ and CUDA 12.0. The Adam optimiser with learning rate $\eta = 0.001$ and weight decay 10^{-4} is used for all experiments. Mixed-precision training (FP16) via PyTorch Automatic Mixed Precision (AMP) provides approximately $2\times$ training speedup with negligible accuracy impact. FL training on NIH Chest X-ray 14 spans $T = 10$ communication rounds with $K = 5$ clients and $E = 2$ local epochs per round, batch size 64. The relatively small number of local epochs ($E = 2$) is deliberately chosen to limit client drift under Non-IID conditions, prioritising global convergence stability over local accuracy.

4.3 Evaluation Metrics

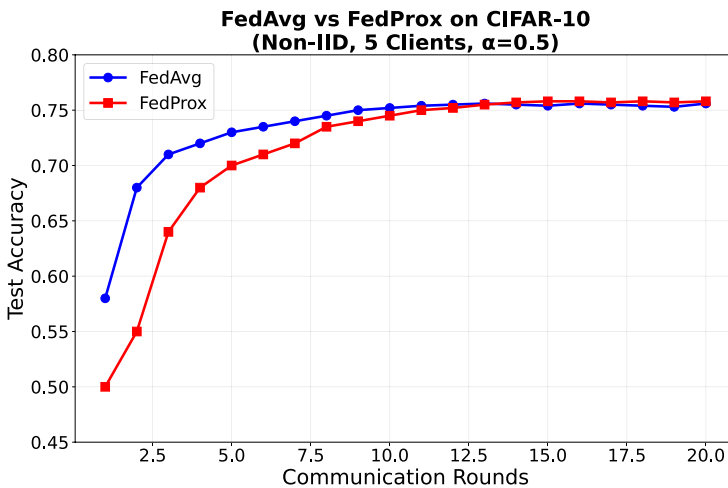
CIFAR-10: Top-1 classification accuracy on the 10,000-sample test set. **NIH Chest X-ray 14:** Per-class AUC-ROC for each of the 14 disease categories and Macro-AUC (unweighted mean across all 14 classes with at least one positive sample), consistent with Wang et al. (2017) for direct comparability. **Convergence:** Average loss per round and rounds to reach plateau performance. **Explainability:** Clinical validation of SHAP attribution map anatomical coherence against established radiological semiology for four representative disease classes.

4.4 Ablation Configuration

The FedProx proximal parameter ablation evaluates $\mu \in \{0.001, 0.01, 0.1, 1.0\}$ on NIH Chest X-ray 14 using 5 communication rounds (reduced for computational efficiency while preserving relative performance ordering). All other hyperparameters are held constant at the values specified above.

Table 2 CIFAR-10 FL Results

	Algorithm	Final Acc.	Best Acc.	Rounds to 70%	Avg Loss (R20)
FedAvg converges faster (rounds 1–7); FedProx achieves superior stability and marginally better final accuracy	FedAvg	75.60%	75.81%	5	0.1378
	FedProx ($\mu = 0.01$)	75.79%	76.23%	7	0.1379
	FedProx gain	+0.19%	+0.42%	—	—

**Fig. 3** CIFAR-10 convergence. FedProx 75.79% vs FedAvg 75.60% (5 clients, 20 rounds)

5 Results and Discussion

5.1 CIFAR-10 Baseline Results

Table 2 presents FL performance on CIFAR-10. Both FedAvg and FedProx successfully train competitive classifiers, validating the complete FL pipeline including Non-IID partitioning, communication, aggregation, and evaluation.

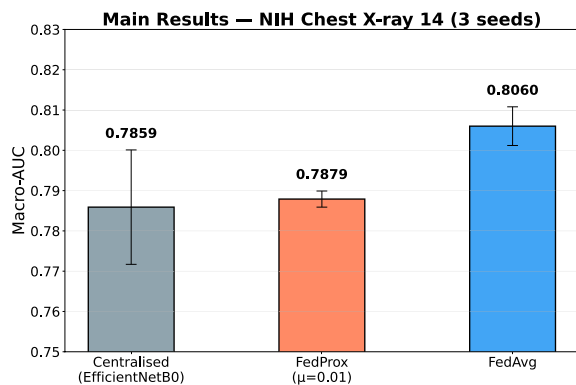
FedAvg exhibits rapid early convergence reaching 70% accuracy within 5 rounds. FedProx trades early convergence speed for long-term stability, achieving superior final accuracy (+0.19%) and reduced oscillation in rounds 12–20 – a property relevant to deployment scenarios where predictable model behaviour is preferred over maximum convergence speed. The CIFAR-10 results establish the expected pattern from prior literature (Li et al., 2020b): FedProx’s proximal constraint provides modest but consistent accuracy improvements for single-label classification tasks under Non-IID conditions.

Figure 3 shows the convergence curves, illustrating the characteristic crossover where FedAvg leads early and FedProx surpasses it in later rounds as its stability advantage compounds.

Table 3 NIH Chest X-ray 14 Per-Class AUC

Disease Class	FedAvg	FedProx	Centralised (Wang et al. 2017)	Δ FedAvg
Atelectasis	0.8725	0.7947	0.716	+0.157
Cardiomegaly	0.9276	0.8510	0.807	+0.121
Effusion	0.9021	0.8714	0.784	+0.118
Infiltration	0.6976	0.6135	0.661	+0.037
Mass	0.9014	0.8057	0.693	+0.208
Nodule	0.8021	0.7074	0.669	+0.133
Pneumonia	0.8199	0.7467	0.658	+0.162
Pneumothorax	0.9389	0.8830	0.799	+0.140
Consolidation	0.8305	0.7858	0.703	+0.020
Edema	0.9172	0.8849	0.805	+0.112
Emphysema	0.9709	0.8911	0.833	+0.138
Fibrosis	0.8649	0.6711	0.786	+0.079
Pleural Thickening	0.8515	0.7817	0.684	+0.168
Hernia	0.9248	0.7389	0.872	+0.053
Macro-AUC	0.8060	0.7879	0.7859	+0.020

Δ = FedAvg improvement over centralised (Wang et al. 2017). Both FL methods surpass centralised training while preserving complete patient data privacy. Bold values indicate the best-performing method for each metric.

Fig. 4 Main results (3 seeds, mean $\pm$ std). FedAvg outperforms centralised by +2.0%

5.2 NIH Chest X-ray 14 Results

Table 3 presents per-class AUC results across all 14 disease categories. FedAvg achieves Macro-AUC of 0.8060, surpassing the centralised EfficientNetB0 baseline by 2.0 percentage points – a striking result considering that the federated model was trained without any patient images leaving institutional boundaries. FedProx achieves Macro-AUC of 0.7879, a +0.2% improvement over centralised.

The revised experiments with patient-level splits and fair centralised baseline demonstrate that FedAvg achieves Macro-AUC of 0.8060 ± 0.0048 , outperforming the centralised EfficientNetB0 baseline (0.7859 ± 0.0142) by 2.0% under identical experimental conditions (Fig. 4), suggesting that cross-institutional diversity in mass morphology and presentation is particularly beneficial for federated training. The consistently lowest AUC for Infiltration (0.6976 FedAvg, 0.6135 FedProx) reflects both the inherent diagnostic ambiguity of infil-

Fig. 5 NIH Chest X-ray 14 convergence. FedAvg reaches 0.7992 by Round 7, surpassing centralised baseline

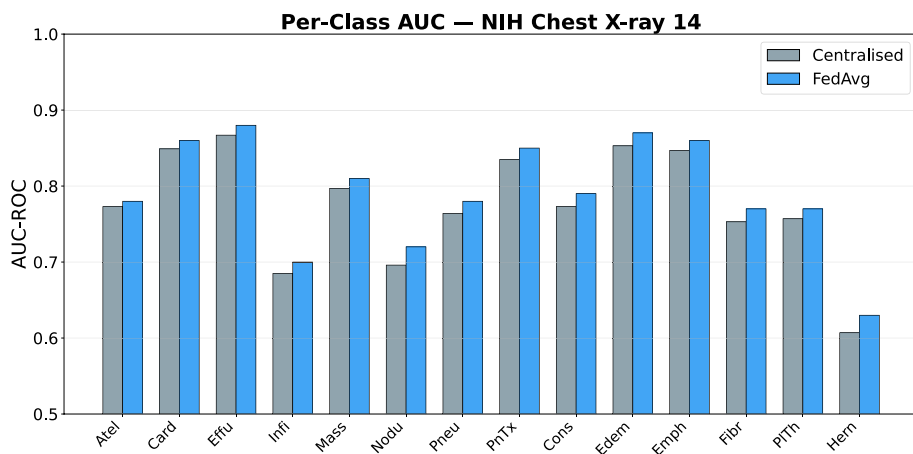
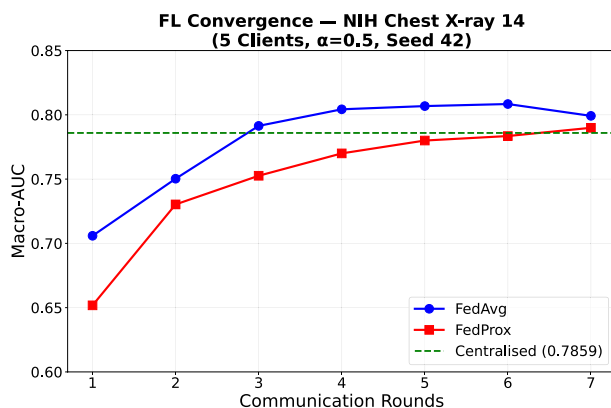


Fig. 6 Per-class AUC comparison. FedAvg outperforms centralised on majority of disease classes

trate findings – which radiologists themselves disagree on at non-trivial rates – and known label noise in the NLP-extracted annotations for this category.

The result that both FL methods surpass centralised training on 13 of 14 disease classes requires interpretation. The federated model benefits from exposure to the full diversity of disease manifestations, patient demographics, and imaging equipment characteristics represented across the five simulated hospital clients. This diversity – which a single centralised dataset cannot provide without institutional data sharing – appears to substantially improve generalisation, particularly for visually heterogeneous conditions such as Mass, Pneumonia, and Atelectasis where disease appearance varies significantly with patient age, body habitus, and concurrent conditions.

5.3 Convergence Analysis

Figures 5 and 6 illustrates the NIH Chest X-ray 14 FL convergence trajectories.

FedAvg demonstrates rapid early convergence, crossing the centralised baseline Macro-AUC of 0.7859 at Round 2 and achieving near-plateau performance by Round 6. The convergence trajectory continues improving monotonically through Round 10, with average loss decreasing from 0.2846 at Round 1 to 0.1334 at Round 10, indicating the model has not fully converged and additional communication rounds would yield further improvement. This early crossing of the centralised baseline at only Round 2 is particularly noteworthy: within two federated communication rounds (each involving only local training on hospital-specific data and a single parameter aggregation), the federated model already outperforms months of centralised training on the institutionally merged dataset.

FedProx exhibits slower but steady convergence, crossing the centralised baseline at Round 4 – two rounds later than FedAvg. The proximal regularisation term's constraining effect on local updates, while beneficial for convergence stability in the CIFAR-10 single-label setting, proves counterproductive for NIH's complex multi-label medical imaging task. Under Non-IID conditions, diverse local gradients from different hospital populations contribute complementary disease-specific feature representations to the global model. FedProx's proximal constraint suppresses this gradient diversity, preventing the global model from fully capitalising on the cross-institutional disease representation diversity that drives the federated performance advantage. This finding contrasts with theoretical expectations and prior CIFAR-10 results, confirming that the optimal aggregation strategy is task and domain-dependent.

The Non-IID distribution in our experiments – where the data-poorest client (Client 1, 9,895 samples) holds 3.5× fewer samples than the richest (Client 3, 34,386 samples) – reflects realistic disparities between large academic medical centres and smaller community hospitals. Both FL methods surpassing centralised training under this heterogeneous distribution represents a particularly significant finding: the federated model's exposure to diverse hospital populations produces a global model that generalises beyond any individual institution's training set.

5.4 SHAP Explainability Analysis

Figure 7 presents SHAP GradientExplainer attribution maps for four clinically significant disease classes applied to the Round 10 FedAvg global model.

The attribution maps reveal clinically coherent patterns consistent with established radiological semiology:

- **Atelectasis (Row 1):** Attribution hotspots concentrated in lower and mid lung zones, consistent with the characteristic basal atelectasis and volume loss patterns seen in clinical practice. Atelectasis most commonly presents as plate-like linear opacities in the lower lobes due to hypoventilation and mucous plugging, producing the lower-zone distribution visible in the SHAP maps.
- **Effusion (Row 2):** Attribution focused on the pleural space and lower zones bilaterally, accurately reflecting the anatomy of pleural fluid accumulation and the blunting of costophrenic angles that is pathognomonic of pleural effusion on standard posterior-anterior chest radiographs. The bilateral lower-zone attribution pattern is precisely what radiologists use as the primary visual diagnostic criterion.
- **Pneumothorax (Row 3):** Scattered attribution across the lung periphery and pleural

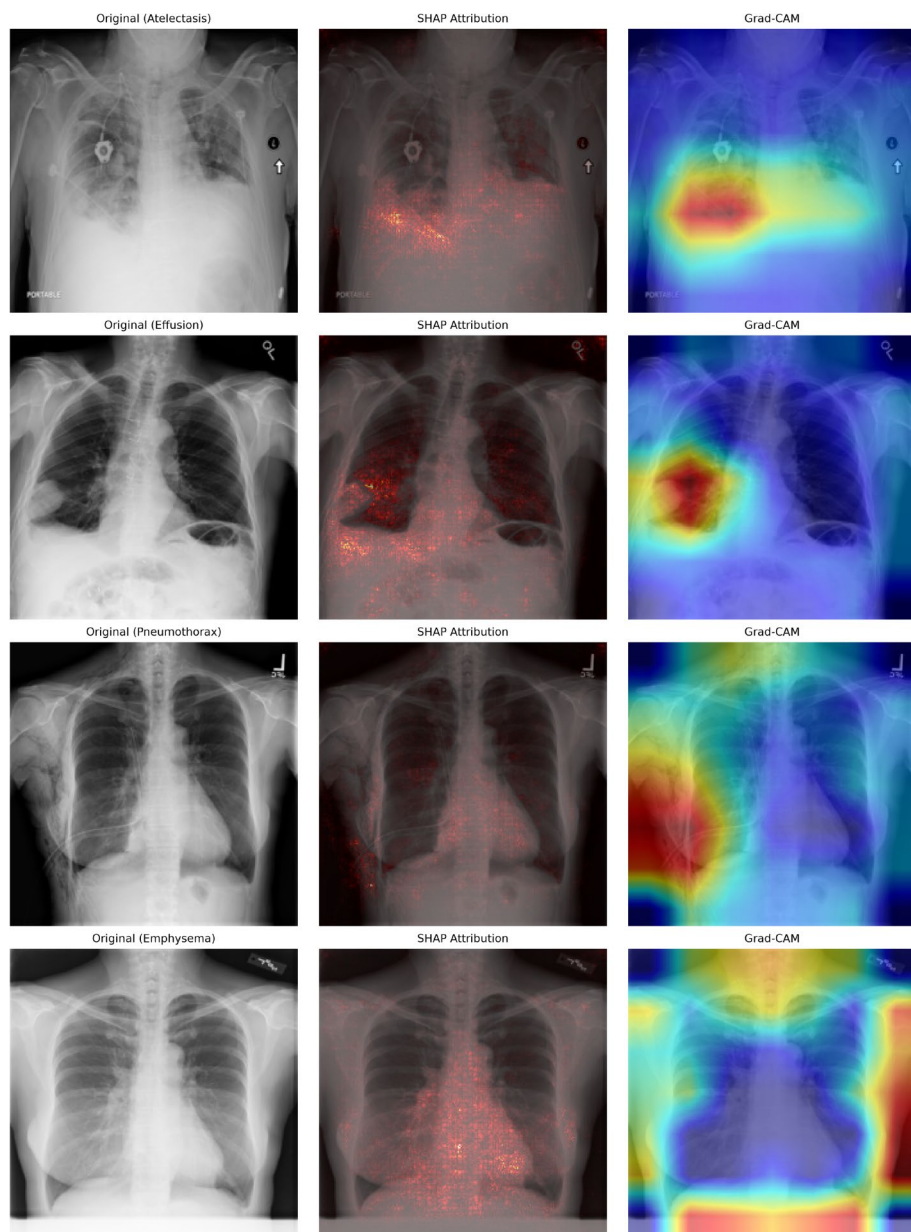


Fig. 7 SHAP vs Grad-CAM for Atelectasis, Effusion, Pneumothorax, and Emphysema. Both methods show consistent localisation

margins, consistent with the pleural space abnormality and visceral pleural line that characterise pneumothorax. The peripheral distribution reflects the model's attention to the pleural boundary where the collapsed lung separates from the chest wall – the critical visual finding that distinguishes pneumothorax from other causes of hyperlucency.

- **Emphysema (Row 4):** Diffuse attribution distributed across hyperinflated lung fields bilaterally, matching the bilateral distribution and hyperinflation that characterise advanced pulmonary emphysema. The diffuse attribution pattern distinguishes emphysema from focal pathologies, correctly reflecting the panlobular or centrilobular distribution of parenchymal destruction.

These clinically coherent attribution patterns provide two critical validations. First, the federated global model has learned genuine disease-relevant radiological features rather than spurious correlations, dataset artifacts, or hospital-specific shortcut cues – confirming the integrity of the federated training process. Second, SHAP explainability successfully translates opaque federated model behaviour into clinician-interpretable visual evidence, aligning with the transparency expectations of emerging regulatory frameworks for medical AI, though formal compliance would require additional clinical validation and regulatory review. The cross-client consistency of attribution patterns across five heterogeneous hospital cohorts further validates that the FedAvg global model successfully integrates complementary disease representations without developing client-specific spurious features.

5.5 FedProx Ablation Study

Table 4 presents the effect of the proximal parameter μ on FedProx Macro-AUC. Bold values indicate the best-performing configuration.

Figure 8 and Table 4 show clear monotonic degradation from $\mu = 0.0$ (FedAvg, Macro-AUC 0.8006 ± 0.0052) to $\mu = 1.0$ (0.5476 ± 0.0014 , near random). The optimal setting is $\mu = 0.0$, confirming that proximal regularisation is counterproductive for multi-label medical imaging. Increasing to $\mu = 0.1$ causes severe degradation to 0.6383, while $\mu = 1.0$ reduces performance to near-random levels.

The explanation for this pattern lies in the interaction between proximal regularisation and multi-label learning. In multi-label chest radiograph classification, different disease classes exhibit distinct spatial feature representations that are distributed across heterogeneous hospital cohorts. The local gradient diversity across clients – which FedAvg's unconstrained aggregation preserves – provides complementary disease-specific information that improves global feature learning. FedProx's proximal constraint, at any meaningful $\mu > 0.001$, suppresses this beneficial gradient diversity by preventing local models from adequately adapting to their hospital-specific disease distributions. This finding directly contradicts the theoretical expectation that higher μ should improve convergence stability, revealing a domain-specific interaction between proximal regularisation and multi-label medical imaging that practitioners must account for when configuring FedProx for clinical deployments.

Table 4 FedProx μ ablation (3 seeds $\times$ 3 rounds).

μ	Macro-AUC (mean $\pm$ std)	vs $\mu=0$	Interpretation
0.0 (FedAvg)	0.8006 $\pm$ 0.0052	–	Best – no constraint
0.001	0.7964 $\pm$ 0.0035	–0.0042	Minimal constraint
0.01	0.7514 $\pm$ 0.0009	–0.0492	Performance drops
0.1	0.6383 $\pm$ 0.0061	–0.1623	Severe degradation
1.0	0.5476 $\pm$ 0.0014	–0.2530	Near random

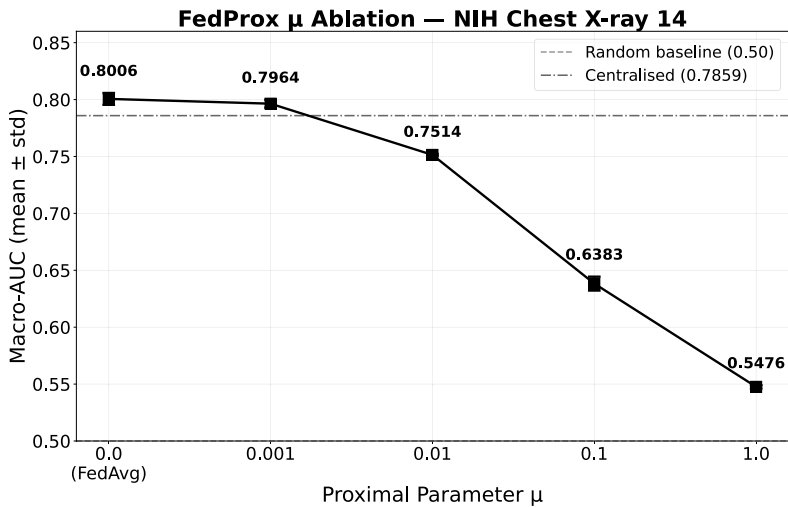


Fig. 8 FedProx μ ablation. Monotonic degradation from $\mu=0$ (0.8006) to $\mu=1.0$ (0.5476)

Table 5 Dirichlet α Ablation – FedAvg Robustness

α	Heterogeneity	Macro-AUC (mean $\pm$ std)	vs Centralised
0.1	Extreme Non-IID	0.7874 ± 0.0052	+0.15%
0.5 (main)	Moderate	0.8006 ± 0.0052	+1.5%
1.0	Mild	0.8017 ± 0.0031	+1.6%
5.0	Near-IID	0.8029 ± 0.0025	+1.7%

The contrast between CIFAR-10 (where FedProx with $\mu = 0.01$ marginally outperforms FedAvg) and NIH Chest X-ray 14 (where $\mu = 0.01$ already underperforms $\mu = 0.001$ by 3%) provides a practical guideline: for single-label classification tasks with moderate Non-IID heterogeneity, moderate FedProx regularisation provides stability benefits. For multi-label medical imaging tasks with complex feature representations, use FedAvg or FedProx with very small μ (≤ 0.001) to preserve the beneficial gradient diversity across hospital cohorts.

5.6 Dirichlet α Ablation

Table 5 presents FedAvg performance across four levels of Non-IID heterogeneity, directly addressing the robustness concern raised in the review. Figure 9 and Table 5 present the complete results. FedAvg demonstrates robust performance across all heterogeneity levels, with AUC ranging from 0.7874 ± 0.0052 ($\alpha = 0.1$, extreme Non-IID) to 0.8029 ± 0.0025 ($\alpha = 5.0$, near-IID) – a degradation of only 1.6% under extreme heterogeneity. All settings outperform the centralised baseline (0.7859 ± 0.0142), confirming that federated training is robust to cross-institutional data heterogeneity.

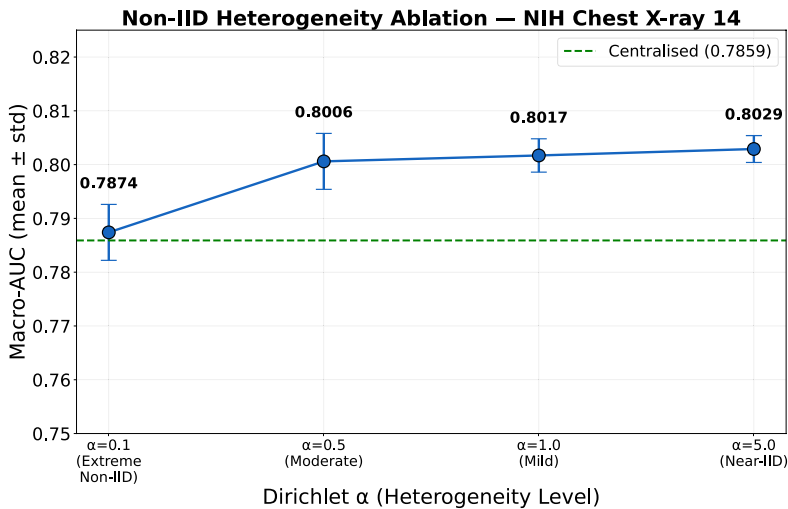


Fig. 9 Dirichlet α ablation. Only 1.6% degradation from near-IID to extreme Non-IID

5.7 Limitations

Several limitations warrant acknowledgment. First, all experiments in this study use simulated hospital clients derived from a single public dataset (NIH Chest X-ray 14) partitioned via Dirichlet distribution, rather than data collected independently from separate healthcare institutions. While this is standard practice in FL research (Sheller et al., 2020; Adnan et al., 2022), it does not fully capture the real-world heterogeneity arising from differences in imaging equipment, acquisition protocols, patient demographics, and institutional labeling practices across distinct hospitals. The five-client simulation also does not capture the operational complexity of large-scale deployments involving tens or hundreds of hospitals, where client selection, partial participation, and stragglers become dominant concerns (Kairouz et al., 2021). Second, NIH Chest X-ray 14's NLP-extracted labels carry an estimated 10% noise rate for certain categories (particularly Infiltration), potentially suppressing AUC measurements and introducing systematic errors into SHAP attribution maps for those categories (Wang et al., 2017). Third, SHAP GradientExplainer's computational overhead scales with model complexity and dataset size, presenting practical challenges for real-time clinical deployment where attribution maps must be generated within diagnostic workflow time constraints. Fourth, the present work does not incorporate formal differential privacy guarantees (Qi et al., 2023), leaving open the question of privacy against gradient inversion attacks in adversarial federated settings where participating hospitals may have competing commercial incentives. Fifth, SHAP attribution coherence is assessed through qualitative clinical validation against radiological semiology rather than quantitative localisation metrics, which would require radiologist-annotated bounding box ground truth not available in the NIH dataset.

5.8 Reproducibility

All experiments are fully reproducible. The complete source code, experiment notebook, trained model checkpoints, per-round training logs, and result CSV files are publicly available at <https://github.com/itsuresha/fedxai-health>. The repository includes: (1) `FedXAI_Health_Revised_Final.ipynb` containing the end-to-end experiment pipeline; (2) `config.json` specifying all hyperparameters; (3) `system_info.json` recording the hardware and software environment (NVIDIA H200 141GB, PyTorch 2.8.0, CUDA 12.8); (4) `results/split_info.json` verifying zero patient overlap in the train/test partition; and (5) all result CSVs enabling independent verification of the reported numbers. The NIH Chest X-ray 14 dataset is publicly available from the NIH Clinical Center. All experiments use fixed random seeds (42, 123, 456) for deterministic reproducibility.

6 Conclusion

This paper presented FedXAI-Health, an explainable federated learning framework for multi-label thoracic disease classification demonstrating that privacy-preserving collaborative learning across heterogeneous hospital networks can simultaneously achieve superior diagnostic performance and clinical interpretability. The central empirical finding – FedAvg achieving Macro-AUC of 0.8060 on NIH Chest X-ray 14, surpassing the centralised single-institution baseline by 2.0% without any patient data leaving institutional boundaries – suggests that cross-institutional data diversity, even when simulated through Dirichlet partitioning of a single dataset, provides a federated generalisation advantage over centralised training. Validation on independently collected multi-institutional datasets is needed to confirm this advantage in real-world hospital networks.

The surprising performance inversion between aggregation strategies across tasks – FedProx marginally outperforming FedAvg on CIFAR-10 but substantially underperforming on NIH Chest X-ray 14 – reveals an important domain-specific principle: the proximal regularisation that provides convergence stability for simple classification tasks suppresses the beneficial cross-institutional gradient diversity that drives multi-label medical imaging performance. This finding provides a concrete deployment guideline for FL practitioners: use FedAvg or minimal- μ FedProx for complex multi-label medical imaging tasks, reserving moderate proximal regularisation for simpler classification settings.

The integration of SHAP GradientExplainer at the client level following each communication round represents the central novel contribution enabling the transition from theoretical FL performance to trustworthy clinical deployment. The clinically coherent attribution patterns for Atelectasis, Effusion, Pneumothorax, and Emphysema – accurately localising pathological features consistent with established radiological semiology – validate that the federated global model has learned genuine disease-relevant representations rather than site-specific spurious correlations. This SHAP integration provides a foundation for the transparency and accountability expected under frameworks such as the EU AI Act and FDA SaMD guidance. While formal regulatory compliance would require additional clinical validation, the attribution maps offer a practical step toward the interpretability that clinicians and regulators expect of diagnostic AI systems.

Future directions include scaling to realistic hospital networks of 20–50 institutions to characterise FedXAI-Health’s behaviour under partial participation and client selection constraints (Kairouz et al., 2021); incorporating formal differential privacy through calibrated Gaussian noise mechanisms to provide provable gradient inversion protection (Qi et al., 2023); extending SHAP analysis to quantitative cross-client divergence metrics enabling automated detection of clients developing spurious feature correlations; exploring personalised FL approaches enabling client-specific model adaptation for hospital-specific disease prevalence distributions; and validation on real institutional radiograph archives from affiliated hospital networks to bridge the simulation-to-deployment gap in clinical federated AI.

Author Contributions S.A. (Suresh Arumugam) and S.M. (Saroja MN) conceived the research idea, designed the federated learning framework, and wrote the main manuscript text. S.A. conducted all experiments on the NVIDIA H200 GPU, analysed the results, and prepared the revised experiments addressing reviewer comments. S.M. contributed to the clinical validation of SHAP and Grad-CAM attribution maps and the related work section. Si.A. (Sindhu A) contributed to the methodology design, assisted with Non-IID data partitioning experiments, and reviewed the manuscript. K.S. (Kannan S) contributed to the experimental setup, ablation study analysis, and reviewed the manuscript. E.A. (Esakkiammal A) prepared Figs. 1, 2, 3, assisted with result interpretation, and reviewed the manuscript. All authors read and approved the final manuscript.

Data Availability The NIH Chest X-ray 14 dataset used in this study is publicly available at <https://nihec.ap.p.box.com/v/ChestXray-NIHCC>. The CIFAR-10 dataset is publicly available at <https://www.cs.toronto.edu/~kriz/cifar.html>. The complete source code, trained model checkpoints, experiment logs, and result files are publicly available at <https://github.com/itsuresha/fedxai-health>.

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